

Mastering Indian Agriculture

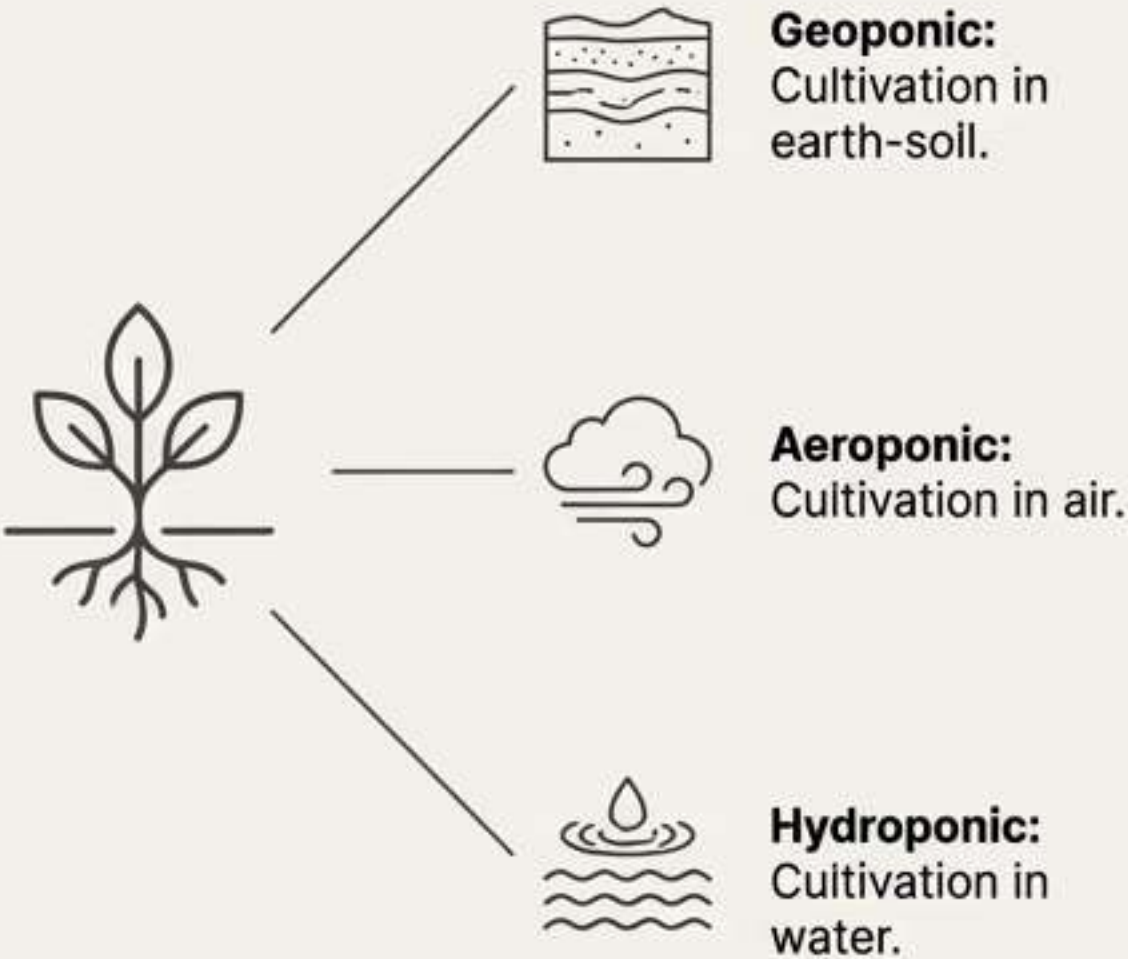
A Strategic Guide for NABARD Grade A & B Aspirants



The Scope of Agriculture: From Soil Cultivation to Social Science

Derived from the Latin words 'ager' (soil) and 'cultra' (cultivation). The art, the science, and the business of producing crops and livestock for man's use and employment.

Main Spheres of Agriculture

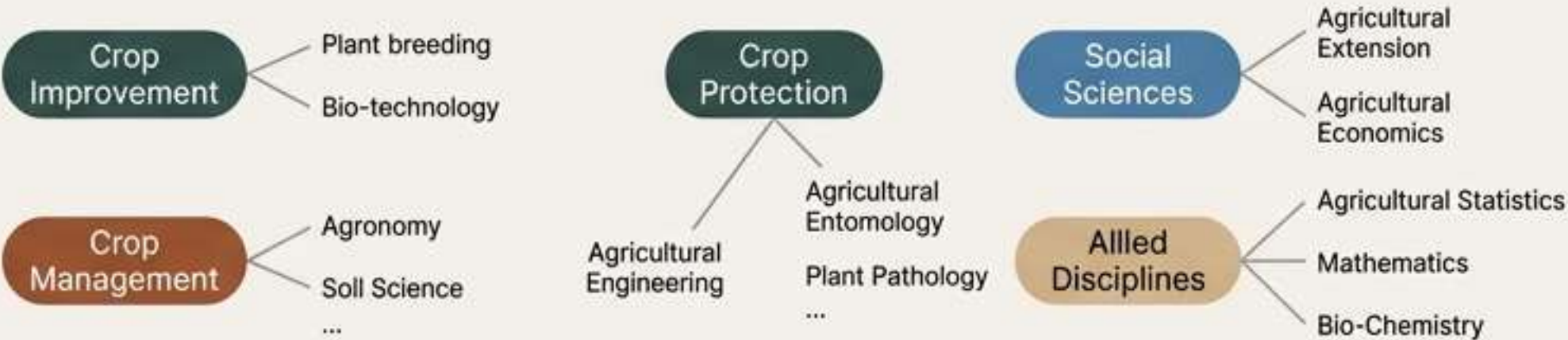


Branches & Allied Disciplines

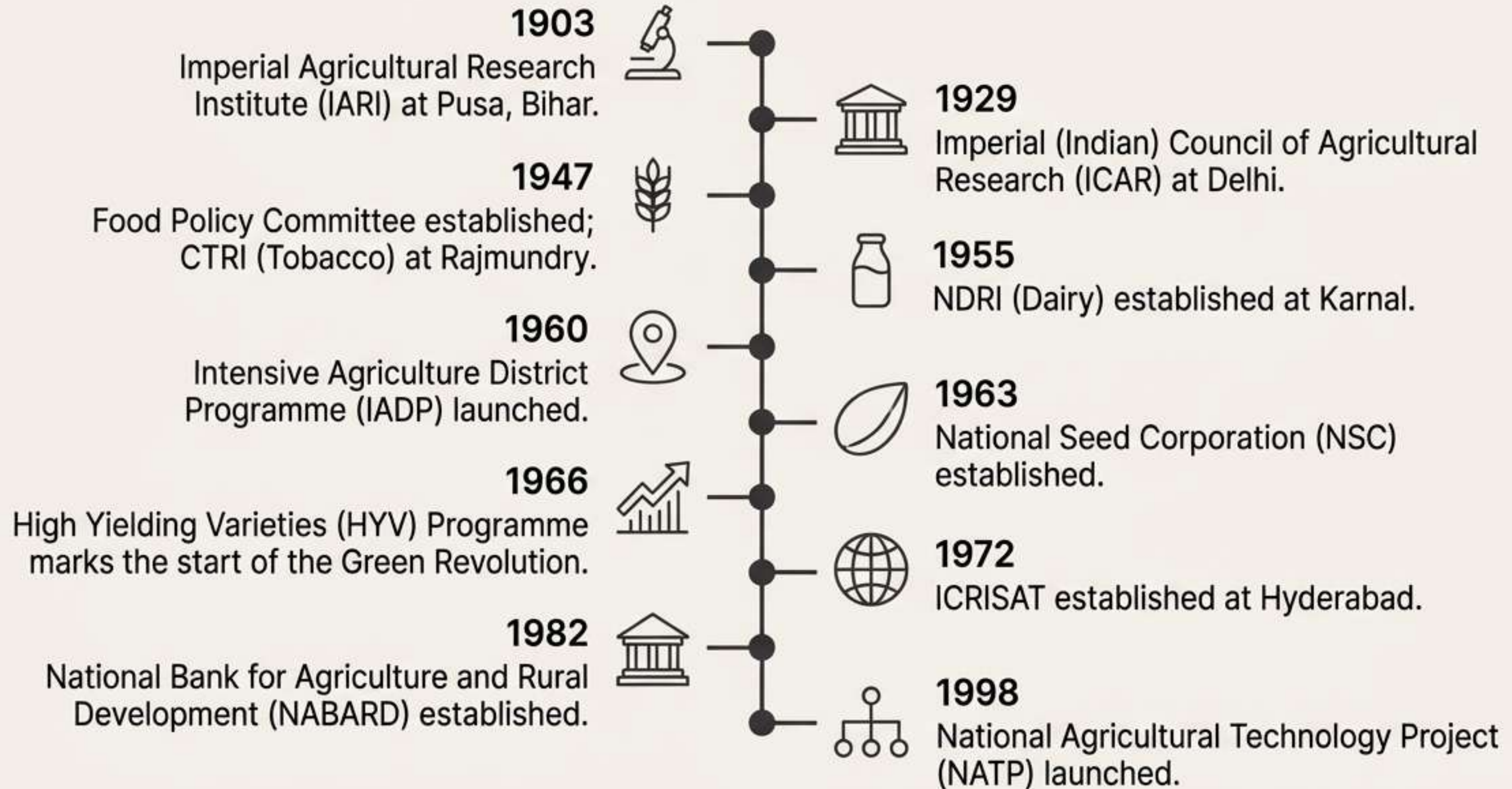
Major Branches



Grouped Disciplines



A Century of Progress: Key Milestones in Indian Agriculture



The Institutional Framework of Agricultural Research

National Institutes

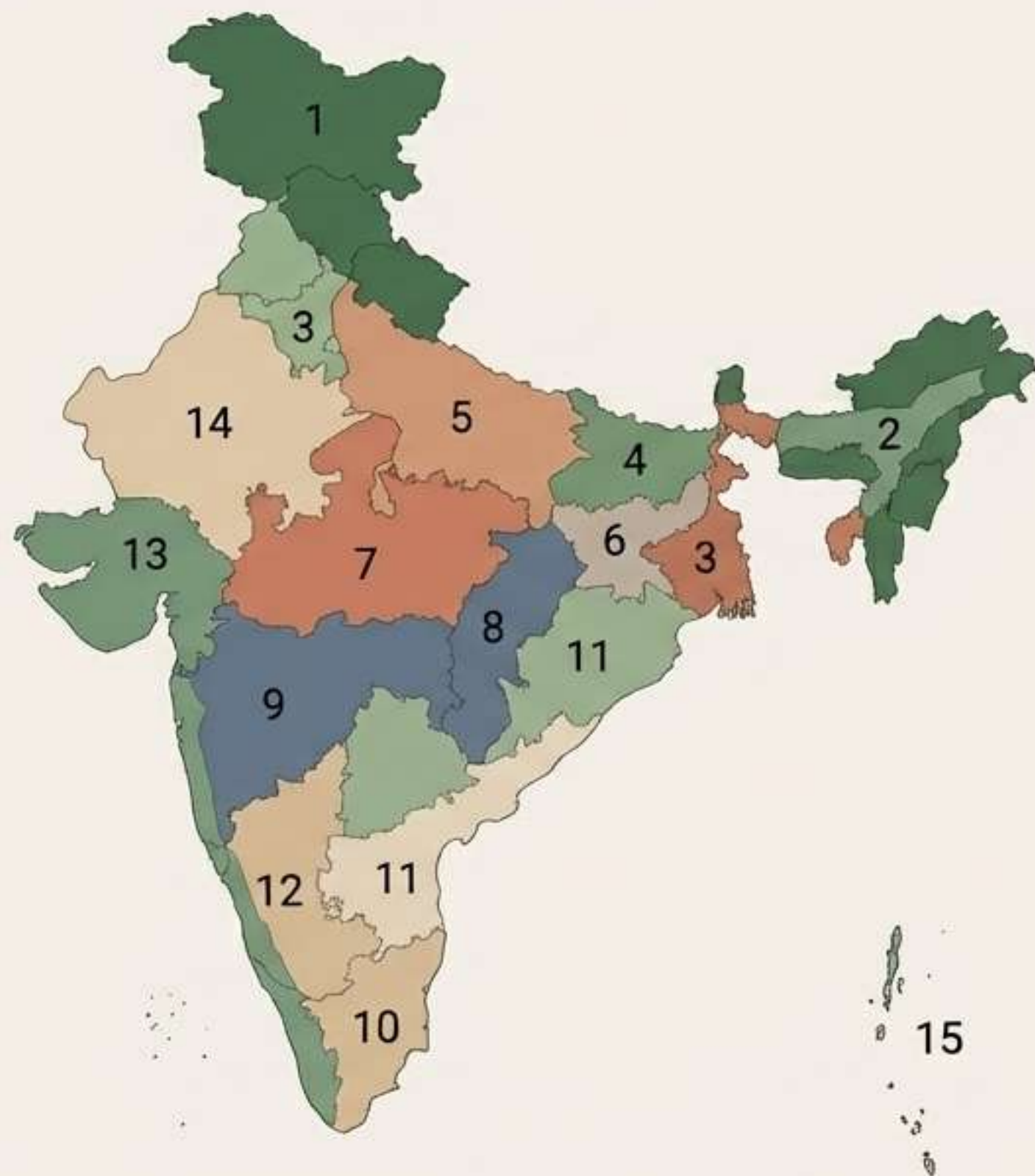
ICAR: Indian Council of Agricultural Research, New Delhi
IARI: Indian Agriculture Research Institute, Pusa, New Delhi
NDRI: National Dairy Research Institute, Karnal, Haryana
CAZRI: Central Arid Zone Research Institute, Jodhpur, Rajasthan
CRRI: Central Rice Research Institute, Cuttack, Orissa
CRIDA: Central Research Institute for Dryland Agriculture, Hyderabad, A.P.
CSSRI: Central Soil Salinity Research Institute, Karnal, Haryana
NBSSLUP: National Bureau of Soil Survey and Land Use Planning, Nagpur, Maharashtra
NAARM: National Academy of Agricultural Research Management, Hyderabad

International Institutes (Located in India)

ICRISAT: International Crops Research Institute for Semi-Arid Tropics, Hyderabad, India
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Agricultural research is carried out by ICAR research centers and State Agricultural Universities (SAUs).

Mapping India's Agricultural Landscape: The 15 Agro-Climatic Zones



Based on homogeneity in agro-characteristics such as rainfall, temperature, soil, topography, cropping systems, and water resources, India is divided into 15 agro-climatic regions.

- | | |
|------------------------------|---------------------------------|
| 1. Western Himalayan Region | 8. Central Plateau and Hills |
| 2. Eastern Himalayan Region | 9. Western Plateau and Hills |
| 3. Lower Gangetic Plains | 10. Southern Plateau and Hills |
| 4. Middle Gangetic Plains | 11. East Coast Plains and Hills |
| 5. Upper Gangetic Plains | 12. West Coast Plains and Ghats |
| 6. Trans-Gangetic Plains | 13. Gujarat Plains and Hills |
| 7. Eastern Plateau and Hills | 14. Western Dry Region |
| | 15. The Islands Region |

A Closer Look: Profiles of Key Agro-Climatic Zones (Part 1)

Western Himalayan Region



States: J&K, Himachal Pradesh, UP Hills.

Characteristics: Steep slopes, erosion-prone, skeletal/podsolic soils.

Crops: Rice, maize, wheat, barley, ginger, saffron, temperate fruits.

Strategy: Land use planning based on slope (agriculture <30%, horticulture 30-50%, forestry >50%).

Eastern Himalayan Region



States: North-East states, Darjeeling hills.

Characteristics: High rainfall, high forest cover, shifting cultivation (Jhum).

Challenge: Soil degradation and erosion from Jhum.

Strategy: Develop high potential for forestry and horticulture with input supply and marketing support.

Trans-Gangetic Plains



States: Punjab, Haryana, Delhi, Chandigarh, Sriganganagar (RJ).

Characteristics: Highest net sown area, highest irrigated area, high cropping intensity (170%), rice-wheat system.

Strategy: Diversify cropping, develop food processing industries for fruits/vegetables.

Western Dry Region



States: 9 districts of Rajasthan.

Characteristics: Hot sandy desert, erratic rainfall (<395mm), deep/brackish groundwater, high land-man ratio.

Crops: Pearl millet, cluster bean, moth (Kharif); Wheat, gram (Rabi).

Strategy: Silvipastoral systems, energy plantations, utilize Indira Gandhi Nahar Project.

A Closer Look: Profiles of Key Agro-Climatic Zones (Part 2)

Southern Plateau and Hills



States: 35 districts of AP, Karnataka, Tamil Nadu.

Characteristics: Typically semi-arid, 81% rainfed farming, low-value cereals and millets dominate.

Strategy: Proven dryland technology in watershed areas, crop diversification to less moisture-requiring crops.

East Coast Plains and Hills



States: Coastal Orissa, AP, Tamil Nadu.

Characteristics: Rice and groundnut are key crops, 70% non-irrigated. Prone to water logging and salinity.

Strategy: Watershed management, desilting tanks, drainage programs, develop brackish water fisheries and aquaculture.

West Coast Plains and Ghats



States: Coastal parts of TN, Kerala, Karnataka, Maharashtra, Goa.

Characteristics: High literacy & unemployment (Kerala). Important for plantation crops (spices, coconut) and fisheries.

Strategy: Diversification to horticulture (Mango, Banana), multi-storied cropping, reclaim saline (khar) and acidic (pokhali) lands.

The Islands Region



States: Andaman & Nicobar, Lakshadweep.

Characteristics: Equatorial, high rainfall (3000mm), largely forest zone, heavy soil loss due to runoff.

Crops: Coconut dominates (nearly half of cropped area).

The Foundation of Farming: A Guide to India's Major Soil Types

Alluvial Soil

Location: Indo-Ganga-Brahmaputra Plains, Coastal Plains.

 **Characteristics:** Highly fertile, formed by river deposits. Two types: Old (Khadar) and New (Bangar).

 **Crops:** Rice, Wheat, Sugarcane, Jute, Oilseeds.

Red Soil

 **Location:** Tamil Nadu, SE Karnataka, Jharkhand, Orissa.

 **Characteristics:** Develops on granite/gneiss rocks, low rainfall. Friable, medium fertility.

 **Crops:** Wheat, Rice, Millets, Pulses (with irrigation).

Desert Soil

Location: Rajasthan, Haryana, South Punjab.

Characteristics: Sandy and saline.

Crops: Bajra, Wheat, Groundnut (with irrigation).



Black (Regur) Soil

 **Location:** Lava Plateaus of Maharashtra, Gujarat, MP.

 **Characteristics:** Clayey, high water-holding capacity, high lime content. "Black Cotton Soil".

 Crops: Cotton, Sugarcane, Jowar, Wheat.

Laterite Soil

 **Location:** Western Ghats, parts of Assam, TN, WB.

 **Characteristics:** Formed under high temp & rainfall. Generally infertile.

 **Crops:** Plantation crops like Tea, Coffee, Coconut, Cashew.

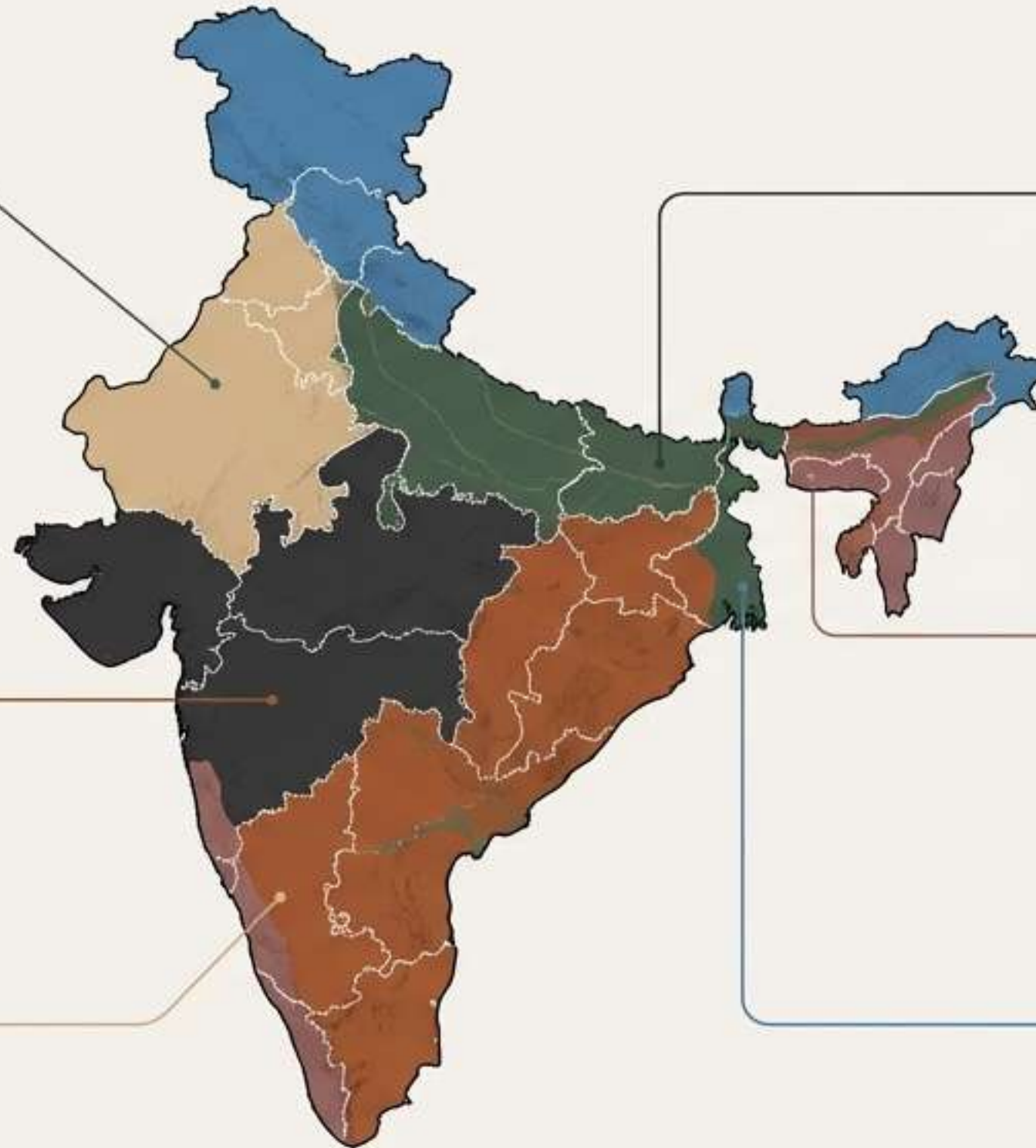
Mountain Soil



Location: Himalayan region.

Characteristics: Varied by altitude
(Alluvium in valleys, Podzols higher up).

Crops: Potatoes, Fruits, Tea, Coffee, Spices.



Maximizing Land Use: Understanding Cropping Systems

Cropping Pattern

The proportion of area under various crops at a point in time.

Cropping System

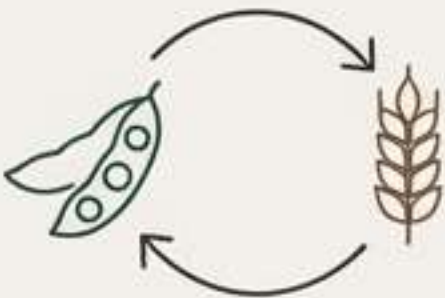
The order and arrangement of crops cultivated on a piece of land over a fixed period.

Monocropping



*Disadvantage: Improper nutrient use, pest/weed buildup.

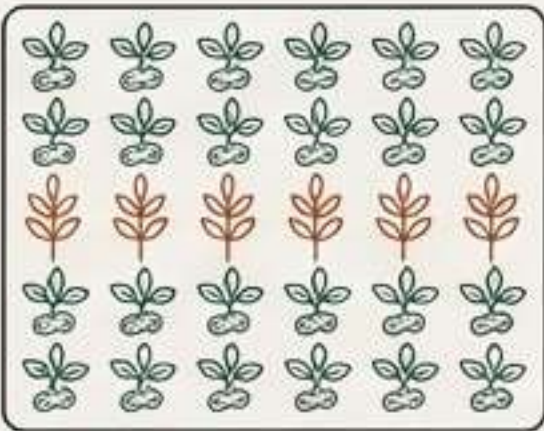
Crop Rotation



*Principle: Tap-rooted crops followed by fibrous-rooted; Legumes after non-legumes to restore fertility.

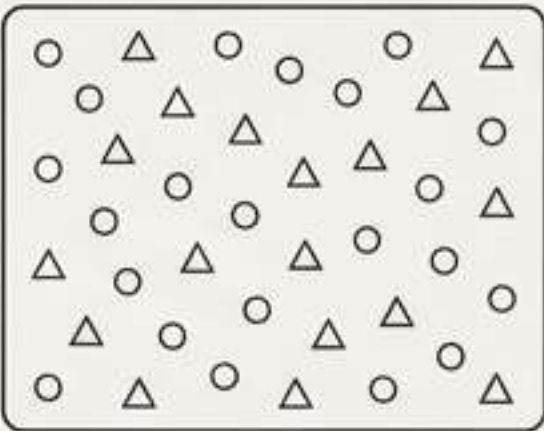
Multiple Cropping

Intercropping



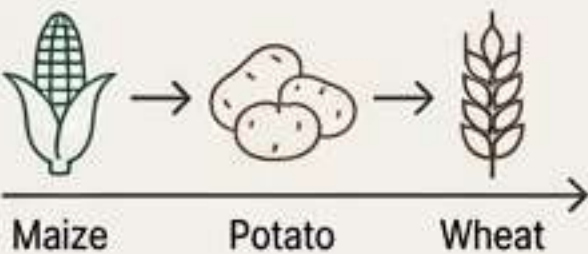
Growing two crops in a definite row pattern.
*Objective: Utilize space between rows.

Mixed Cropping



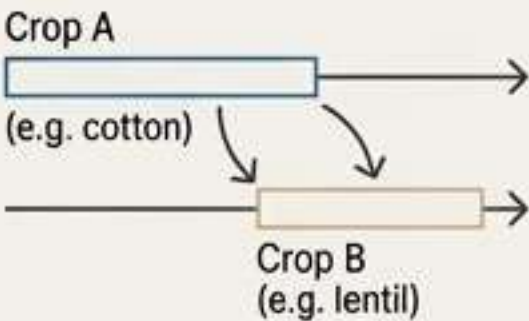
Growing two crops sown together without a fixed pattern.
*Objective: Mitigate risk from climatic hazards.

Sequence Cropping



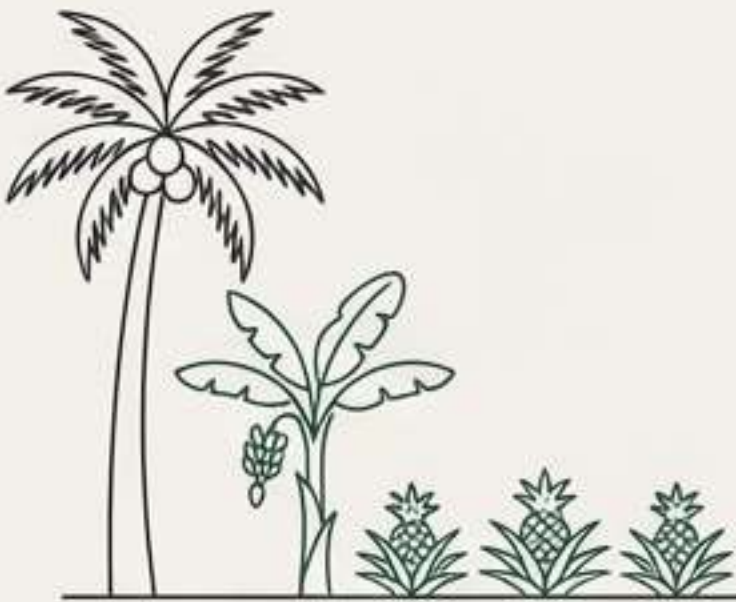
Growing two or more crops in sequence in one year.

Relay Cropping



Sowing the succeeding crop before harvesting the preceding one.

Multi-Storeyed System



Growing plants of different heights simultaneously.

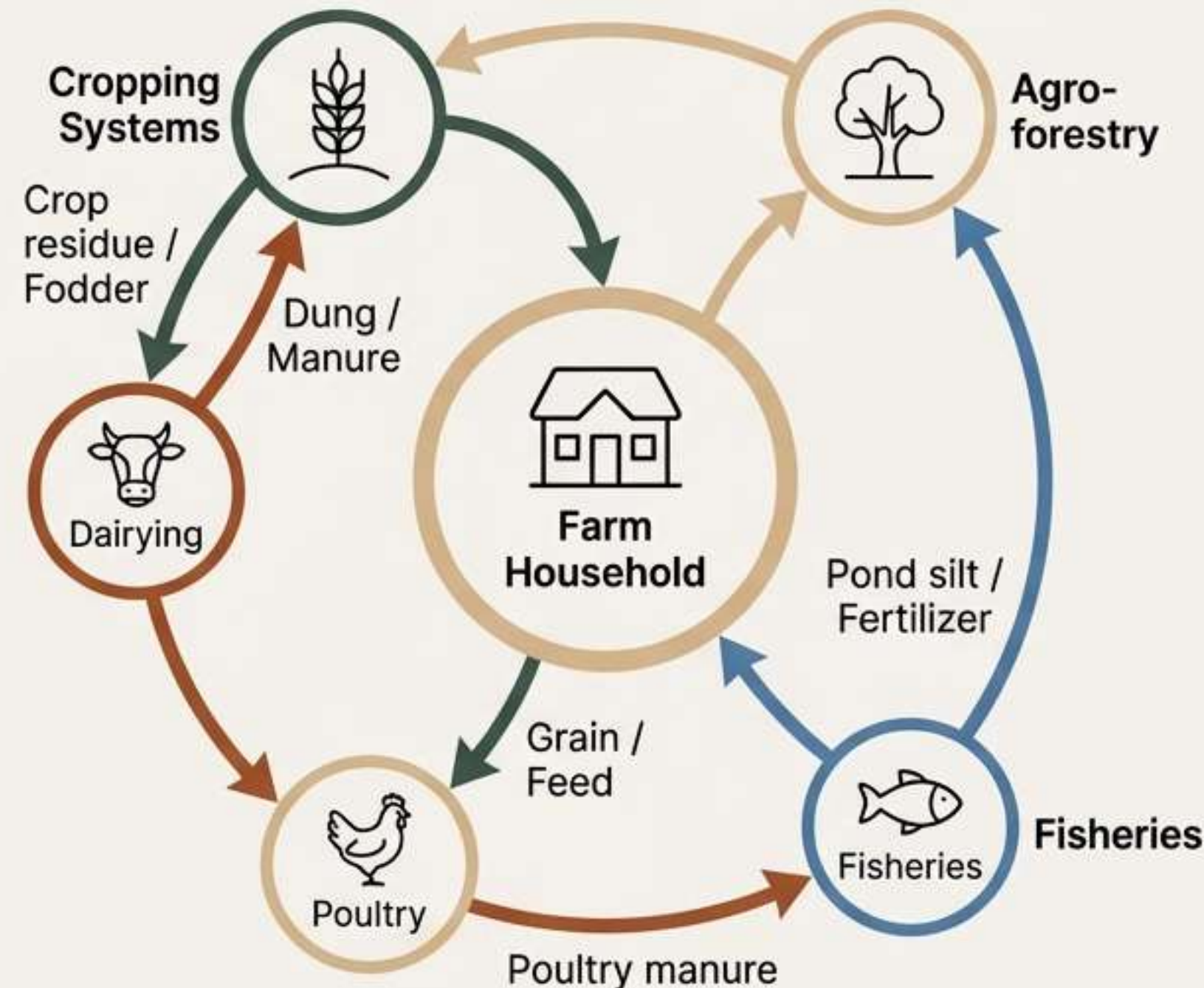
The Integrated Farm: A Holistic System for Sustainability and Profitability

A resource management strategy to achieve economic and sustained production by integrating farm enterprises like cropping, animal husbandry, fisheries, forestry, etc., for optimal resource utilization.

The farm is viewed in a holistic manner, where the waste of one enterprise becomes the input for another.

Key Principles

- ✓ Minimization of risk
- ✓ Recycling of wastes and residues
- ✓ Optimum utilization of resources
- ✓ Maximum productivity and profitability
- ✓ Generation of employment
- ✓ Ecological balance



Key Objectives

- 🎯 Profitability
- 🎯 Potentiality (Soil Health)
- 🎯 Balanced Food Production
- 🎯 Environmental Safety
- 🎯 Round-the-year cash flow

Organic Farming: A System for Agro-Ecosystem Health

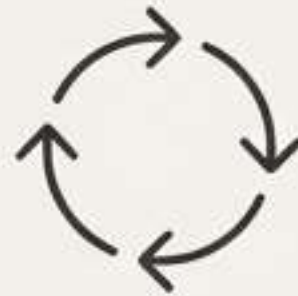
A holistic production management system which promotes and enhances agro-ecosystem health, including biodiversity, biological cycles, and soil biological activity. It avoids or largely excludes the use of synthetically compounded fertilizers, pesticides, and growth regulators



Produce high-quality, nutritious food.



Maintain and increase long-term soil fertility.



Work within a closed system for organic matter and nutrients.



Help in the conservation of soil and water.



Use renewable resources in locally organized systems.



Maintain genetic diversity and protect wildlife habitats.

A Solution to Modern Farming Challenges

- Degradation of soil quality.
- Pollution of soil, water, and food with pesticides and nitrates.
- Development of pesticide resistance in pests.
- Dependence on costly off-farm inputs.

Key Components

- Diverse crop rotations
- Soil fertility management (manures, green manuring, bio-fertilizers)
- Natural pest and disease control
- Weed control through mechanical and biological methods

Cultivating Drylands: Strategies for Water-Scarce Agriculture

Constituent	Dry Farming	Dryland Farming	Rainfed Farming
Rainfall (mm)	< 750 mm	> 750 mm	> 1150 mm
Growing Season	< 75 days	75 - 120 days	> 120 days
Region Type	Arid	Semi-arid	Humid
Moisture	Shortage is common	Prolonged dry spells	Not a major constraint
Key Challenge	Moisture conservation	Moisture conservation & drainage	Disposal of excess water

The Importance of Dry Farming in India

60%



of cultivated area is under dryland agriculture.

40%



of India's human population is supported by it.

60%



of India's livestock depends on it.

75% 90%



Produces 75% of oilseeds and 90% of pulses.

To achieve an evergreen revolution, we must make these 'grey areas' green through technological innovation and resource conservation.

Soil Health: Understanding Fertility, Productivity, and the Threat of Erosion

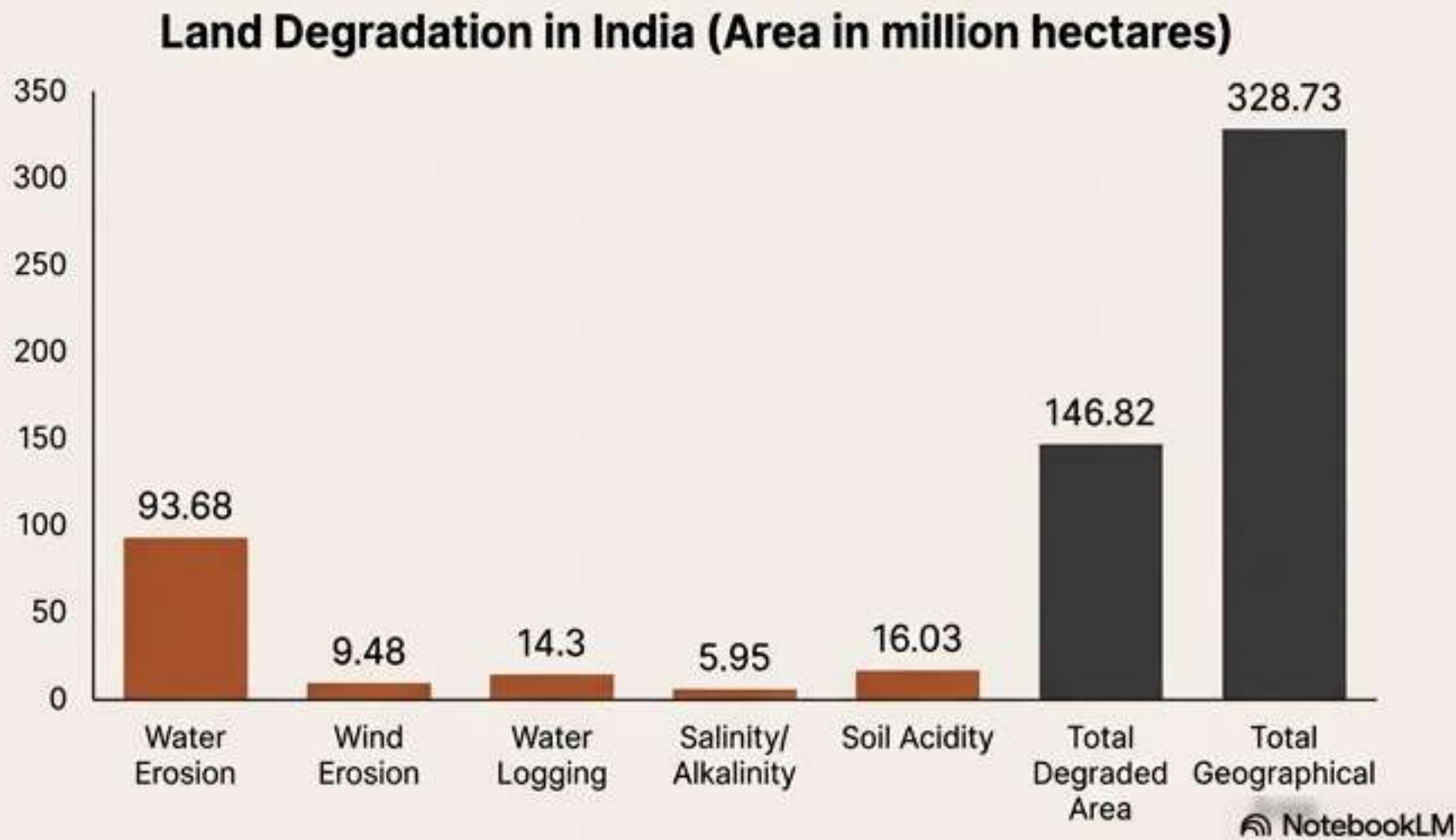
Soil Fertility vs. Soil Productivity

Soil Fertility	Soil Productivity
The inherent capacity of soil to supply plant nutrients.	The capacity of soil to produce crops , expressed in yield.
An aspect of the soil-plant relationship.	An economic concept , influenced by fertility, management, and climate.
A fertile soil is not always productive (e.g,a fertile soil in a desert).	A productive soil is always fertile.

The Scale of Soil Erosion in India

"Soil erosion is the detachment, transport, and deposition of soil particles from the land surface."

It causes both on-site (loss of productivity) and off-site (siltation, pollution) problems.



Protecting Our Foundation: Soil & Water Conservation Practices

Biological Measures (Agronomic & Agrostological)

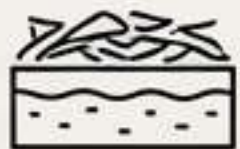
"Working with Nature"



Contour Farming: Plowing and planting across the slope, not up and down.



Crop Rotation & Mixed Cropping: Alternating crops (especially legumes) to improve soil structure and fertility.



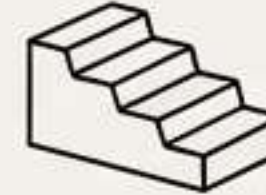
Mulching & Cover Crops: Covering the soil with crop residue or planting cover crops to protect from raindrop impact.



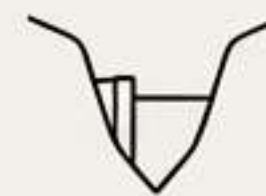
Afforestation & Ley Farming: Planting trees (windbreaks, shelterbelts) and rotating crops with grasses to bind the soil.

Mechanical Measures (Engineering)

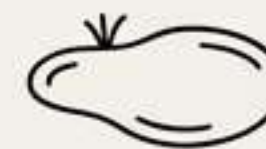
"Engineering the Landscape"



Contour Terracing & Bunding: Creating step-like benches or ridges on slopes to slow runoff.



Gully Control: Using structures like check dams (brush, earth, concrete) to stop gullies from expanding.



Ponds & Reservoirs: Constructing small ponds to harvest runoff water.



Stream Bank Protection: Stabilizing river banks with vegetation or structures (pitching).

These practices are often integrated within a **Watershed Management** approach, which aims for the sustainable use of all resources within a watershed area.

The Critical First Step: Ensuring Quality Through Seed Production

"Reap as you sow." The quality of the seed is the most basic and critical input for successful crop production.



Genetic Purity:
True to type.



Physical Purity:
Free from other seeds, weeds, and inert matter.



High Germination:
Assured germination percentage and vigorous seedlings.



Health: Uniform, well-developed, and free from pathogens.



Moisture:
Optimal moisture content (12-14%).

Breeder Seed (100% Purity)

Produced by the original plant breeder/institution. Source for Foundation Seed.



Foundation Seed

Progeny of Breeder Seed. Produced on government farms or by specialized growers under supervision of the National Seeds Corporation (NSC).



Certified Seed

Progeny of Foundation Seed. Produced by progressive farmers according to standards set by State Seed Certification Agencies. This is the seed available to farmers for commercial crop production.

Key Organization

The **National Seeds Corporation (NSC)**, established in 1963, is responsible for promoting the seed industry and the production/supply of foundation seeds.